STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-DERIVED-PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4551-4563 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4565-4619, the complete author proof (53 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_sdga.json (pb=2234, stmt 4551-4563, proof 4565-4619).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-derived-products' also exists in stacks-dga — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FRJ: sources/stacks-sdga/pinned-original-source.tex, source lines 1444–1455. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FT8: sources/stacks-sdga/pinned-original-source.tex, source lines 4508–4520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FS2: sources/stacks-sdga/pinned-original-source.tex, source lines 2535–2541. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSV: sources/stacks-sdga/pinned-original-source.tex, source lines 3968–3976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSR: sources/stacks-sdga/pinned-original-source.tex, source lines 3776–3784. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FT0: sources/stacks-sdga/pinned-original-source.tex, source lines 4233–4242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Then*

1. *$D(\mathcal{A}, d)$ has both direct sums and products,*
2. *direct sums are obtained by taking direct sums of differential graded $\mathcal{A}$-modules,*
3. *products are obtained by taking products of K-injective differential graded modules.*

Proof. We will use that $Mod(\mathcal{A}, \mathrm{d})$ is an abelian category with arbitrary direct sums and products, and that these give rise to direct sums and products in $K(Mod(\mathcal{A}, \mathrm{d}))$. See Lemmas 0FRJ and 0FS2.

Let $\mathcal{M}_j$ be a family of differential graded $\mathcal{A}$-modules. Consider the direct sum $\mathcal{M} = \bigoplus \mathcal{M}_j$ as a differential graded $\mathcal{A}$-module. For a differential graded $\mathcal{A}$-module $\mathcal{N}$ choose a quasi-isomorphism $\mathcal{N} \to \mathcal{I}$ where $\mathcal{I}$ is graded injective and K-injective as a differential graded $\mathcal{A}$-module. See Theorem 0FT0. Using Lemma 0FT8 we have

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{M}, \mathcal{N}) &= \mathrm{Hom}_{K(\mathcal{A},\mathrm{d})}(\mathcal{M}, \mathcal{I}) \\ &= \prod \mathrm{Hom}_{K(\mathcal{A},\mathrm{d})}(\mathcal{M}_j, \mathcal{I}) \\ &= \prod \mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{M}_j, \mathcal{I}) \end{aligned}$$

whence the existence of direct sums in $D(A, \mathrm{d})$ as given in part (2) of the lemma.

Let $\mathcal{M}_j$ be a family of differential graded $\mathcal{A}$-modules. For each j choose a quasi-isomorphism $\mathcal{M} \to \mathcal{I}_j$ where $\mathcal{I}_j$ is graded injective and K-injective as a differential graded $\mathcal{A}$-module. Consider the product $\mathcal{I} = \prod \mathcal{I}_j$ of differential graded $\mathcal{A}$-modules. By Lemmas 0FSV and 0FSR we see that $\mathcal{I}$ is graded injective and K-injective as a differential graded $\mathcal{A}$-module. For a differential graded $\mathcal{A}$-module $\mathcal{N}$ using Lemma 0FT8 we have

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{N}, \mathcal{I}) &= \mathrm{Hom}_{K(\mathcal{A},\mathrm{d})}(\mathcal{N}, \mathcal{I}) \\ &= \prod \mathrm{Hom}_{K(\mathcal{A},\mathrm{d})}(\mathcal{N}, \mathcal{I}_j) \\ &= \prod \mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{N}, \mathcal{M}_j) \end{aligned}$$

whence the existence of products in $D(\mathcal{A}, \mathrm{d})$ as given in part (3) of the lemma. □